

Particle Problem Sheet

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1 Introduction

1.1 Charge, baryon number and lepton number conservation

In all particle interactions, charge, baryon number, and lepton number need to be conserved [1].

Take, for example, beta minus decay, where a neutron turns into a proton by converting a down quark into an up quark, electron and electron antineutrino [2].

$$d \rightarrow u + e^- + \bar{\nu}_e \quad (1)$$

A down quark has a charge of $-\frac{1}{3}e$, which balances with the $\frac{2}{3}e$ charge of an up quark summed with the $-1e$ charge of an electron and the neutral charge of the electron antineutrino.

Both the down- and up-quarks have a baryon number of $\frac{1}{3}$ and the electron and electron antineutrino have a baryon number of 0.

The quarks have a lepton number of 0, and the $+1$ lepton number of the electron cancels out with the -1 lepton number of the electron antineutrino.

Beta plus decay is similar to this:

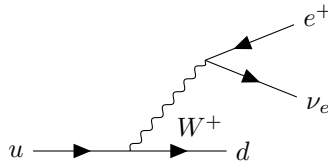


Figure 1: Feynman diagram for β^+ decay.

$$u \rightarrow d + e^+ + \nu_e \quad (2)$$

An up quark has a charge of $\frac{2}{3}e$, which balances with the $\frac{-1}{3}e$ charge of a down quark summed with the $1e$ charge of a positron and the neutral charge of the electron neutrino.

Both the down- and up-quarks have a baryon number of $\frac{1}{3}$ and the positron and electron neutrino have a baryon number of 0.

The quarks have a lepton number of 0, and the -1 lepton number of the positron cancels out with the $+1$ lepton number of the electron neutrino.

This is a particle interaction between hadrons. The baryon and lepton numbers of their fundamental particles are still conserved.

$$\pi^- + p \rightarrow n + \pi^- + \pi^+ \quad (3)$$

This can be simplified (for the purpose of showing conservation) into:

$$p \rightarrow n + \pi^+$$

$$u + u + d \rightarrow (u + d + d) + (u + \bar{d})$$

$$0 \rightarrow d + \bar{d}$$

The baryon number and charge of any particle is the negative of the baryon number and charge of its antiparticle. Therefore the baryon and lepton number is conserved.

A kaon can decay into pi mesons.

$$K^+ \rightarrow \pi^+ + \pi^0 \quad (4)$$

Simplifying once again to show conservation:

$$u + \bar{s} \rightarrow (u + \bar{d}) + (d + \bar{d})$$

$$\bar{s} \rightarrow (\bar{d}) + (d + \bar{d})$$

d and $\bar{d}$ have opposite charge to each other so can be cancelled.

$$\bar{s} \rightarrow (\bar{d})$$

Both the strange quark and down quark have a charge of $\frac{1}{3}$, so charge is conserved.

We can calculate the "quark binding energy" of the kaon.

$$M_K = M_u + M_{\bar{s}} - \frac{B}{c^2}$$

$$B = (M_u + M_{\bar{s}} - M_K) c^2$$

An up quark is 2.16 MeV and a strange quark is 94 MeV [3]. A kaon has a rest mass of 0.526 multiplied by the mass of a proton [2].

$$B = 2.16 + 94 - 0.526 \times 938$$

$$B = -397 \text{ MeV}$$

Similarly, the binding energy of a pi meson is:

$$M_{\pi^+} = M_u + M_{\bar{d}} - \frac{B}{c^2}$$

$$B = (M_u + M_{\bar{d}} - M_{\pi^+}) c^2$$

An up quark is 2.16 MeV and a down quark is 4.7 MeV [3]. A kaon has a rest mass of 0.149 multiplied by the mass of a proton [2].

$$B = (2.16 + 4.7 - 0.149 \times 938$$

$$B = -133 \text{ MeV}$$

The reason kaons and pions have negative "quark binding energies" is because the strong interaction makes the bound state more favourable. Due to the strong force, quarks cannot exist as isolated particles and are confined into hadrons.

A Σ^- baryon is a hadron made of three quarks and undergoes the following decay:[2]

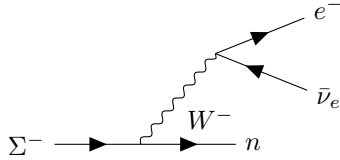


Figure 2: Feynman diagram for $\Sigma^- \rightarrow n + e^- + \bar{\nu}_e$.

$$\Sigma^- \rightarrow n + e^- + \bar{\nu}_e \tag{5}$$

This also follows conservation laws.

$$d + d + s \rightarrow u + d + d + e^- + \bar{\nu}_e$$

$$s \rightarrow u + e^- + \bar{\nu}_e$$

A strange quark has a charge of $-\frac{1}{3}e$, which balances with the $\frac{2}{3}e$ charge of an up quark summed with the $-1e$ charge of an electron and the neutral charge of the electron antineutrino.

Both the strange- and up-quarks have a baryon number of $\frac{1}{3}$ and the electron and electron antineutrino have a baryon number of 0.

The quarks have a lepton number of 0, and the $+1$ lepton number of the electron cancels out with the -1 lepton number of the electron antineutrino.

1.2 Cloud Chambers vs Bubble Chambers

Cloud chambers allow the paths of charged particles to be seen. The top of the chamber is fitted with a paper tissue moistened with isopropanol and uses dry ice. Vapour falls from the soaked tissue and a mist of isopropanol droplets will form just above a cold metal plate (cooled by dry ice). Above this reason the vapour is super-saturated, where a charged particle moving past will cause the vapour to polarise, so the paths of the particles are visible.[4]

A bubble chamber contains a liquefied gas that is rapidly depressurised to create a superheated liquid, allowing passing particles to produce tiny bubbles along their tracks through ionisation. The bubbles are photographed and the resulting tracks are analysed to identify particles and measure their properties.[5]

1.3 Pair Production

If the energy of a gamma ray is high enough, it can produce a matter-antimatter pair.[2] What is the minimum energy required for a gamma ray to produce a pair of muons? We assume that the muons have a kinetic energy of 0. The mass of a muon is $105.66 \text{ MeV}/c^2$. [2]

$$E = \frac{hc}{\lambda}$$

$$E = 2m_{\mu}c^2$$

$$\lambda = \frac{hc}{2m_{\mu}c^2}$$

$$\lambda = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{2 \times 105.66 \times 1.60 \times 10^{-13}}$$

$$\lambda = 5.88 \times 10^{-15} \text{m}$$

What would the energy of a gamma ray be if it produces an electron-positron pair, both moving at $0.8c$? We can use a relativistic formula to determine the energy of the gamma ray.

$$E = (1 - \frac{u^2}{c^2})^{-\frac{1}{2}} 2m_e c^2$$

$$E = (1 - 0.8^2)^{-\frac{1}{2}} \times 2 \times \frac{9.1 \times 10^{-31} \times (3 \times 10^8)^2}{1.60 \times 10^{-13}}$$

$$E = 1.71 \text{ MeV}$$

1.4 Bremsstrahlung

Bremsstrahlung is a German word meaning "breaking radiation". It refers to when a high energy particle (energy much greater than its rest energy) emits electromagnetic radiation when accelerating.

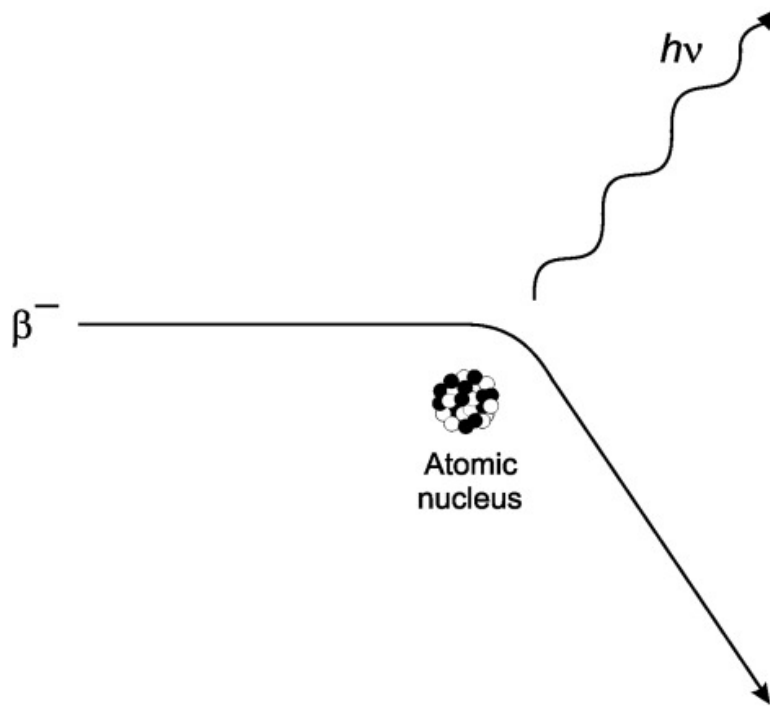


Figure 3: Diagram illustrating Bremsstrahlung [6]

A single high energy electron could result in a "shower" of positrons and electrons. The electron can emit a high energy photon (in the form of a gamma ray), which can in turn produce an electron-positron pair. This process can repeat with the electron produced. The many charged particles can be detected , and we can trace the shower back to its starting point to find the original single high energy electron.[7]

1.5 Koide Formula

The Koide formula connects the masses of particles by summing their masses and dividing them by the square of their sums of their square roots.

$$\text{Koide Mass Ratio} = \frac{m_1 + m_2 + m_3}{(\sqrt{m_1} + \sqrt{m_2} + \sqrt{m_3})^2} \quad (6)$$

$$\text{Koide Mass Ratio} = \frac{m_u + m_d + m_s}{(\sqrt{m_u} + \sqrt{m_d} + \sqrt{m_s})^2}$$

$$\text{Koide Mass Ratio} = \frac{2.16 + 4.7 + 94}{(\sqrt{2.16} + \sqrt{4.7} + \sqrt{94})^2}$$

$$\text{Koide Mass Ratio} = 0.56 \approx \frac{5}{9}$$

2 Strangeness

2.1 Determining the quark content of K^0

The strangeness of a particle is the number of strange quarks in the particle multiplied by -1. The Ω^- particle has a strangeness of -3, and the following particle interaction conserves strangeness, what is the quark content of K^0 ?[2]

$$K^- + p \rightarrow \Omega^- + K^+ + K^0 \quad (7)$$

$$(\bar{u} + s) + (u + u + d) = (s + s + s) + u + \bar{s} + K^0$$

$$K^0 = \bar{s} - d$$

$$\text{Charge} = \bar{s} - d = -\frac{1}{3} - \frac{1}{3} = 0$$

$$\text{Baryon Number} = \bar{s} - d = \frac{1}{3} - \frac{1}{3} = 0$$

Therefore, it must be a meson, as the quark and anti quark pair cancel to give a baryon number of 0.

$$\text{Strangeness} = \bar{s} - d = 1 + 0 = 1$$

The particle has to contain one strange antiquark, which has a charge of $\frac{1}{3}$. The other quark must have a charge of $-\frac{1}{3}$, leaving options of the down and bottom quark. The bottom quark has too much mass compared to the strange antiquark to interact with it.

Using this information, we find that K^0 contains a strange antiquark and a down quark.

2.2 Is Strangeness always conserved?

The following particle interaction has two vertices.

$$\Omega^- \rightarrow \Xi^0 + \pi^- \quad (8)$$

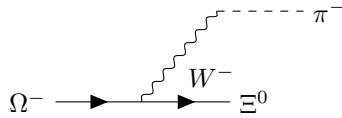


Figure 4: Feynman diagram for $\Omega^- \rightarrow \Xi^0 + \pi^-$.

Vertex 1:

$$s \rightarrow u + W^{-}$$

A strange quark has a charge of $-\frac{1}{3}e$, which balances with the $\frac{2}{3}e$ charge of an up quark summed with the $-1e$ charge of a W minus boson.

Both the strange- and up-quarks have a baryon number of $\frac{1}{3}$ and the W minus boson has a baryon number of 0.

All particles have a strangeness of 0.

Vertex 2:

$$W^{-} \rightarrow d + \bar{u}$$

The W minus boson has a charge of -1 , which balances with the $-\frac{2}{3}e$ charge of an up antiquark summed with the $-\frac{1}{3}e$ charge of a down quark.

The W minus boson has a baryon number of 0, which balances with the $-\frac{1}{3}$ baryon number of an up antiquark summed with the $\frac{1}{3}$ baryon number of a down quark.

The W minus boson has a strangeness of -1, whereas the up antiquark and down quark both have a strangeness of 0, showing that strangeness is not always conserved in weak interactions.[2]

3 Energy and Momentum Calculations

A tau lepton has a very short lifetime and is very likely to decay. Due to its large mass, it decays into hadrons. For example, around 10.8% of the time the following interaction occurs.[2]

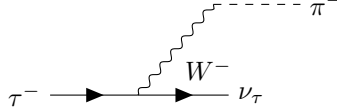


Figure 5: Feynman diagram for $\tau^- \rightarrow \pi^- + \nu_\tau$.

$$\tau^- \rightarrow \pi^- + \bar{\nu}_\tau \quad (9)$$

The tau neutrino is produced to conserve lepton number, as τ^- balances with $\bar{\nu}_\tau$, both having a lepton number of 1.

A tau minus lepton has a mass of $1776.86 \text{ MeV}/c^2$. [2] Assume it is moving at a speed of $0.2c$. We can calculate its kinetic energy

$$E = \left(1 - \frac{u^2}{c^2}\right)^{-\frac{1}{2}} m_{\tau^-} c^2 - m_{\tau^-} c^2$$

$$E = (1 - 0.2^2)^{-\frac{1}{2}} \times 1776.86 - 1776.86$$

$$E = 36.64 \text{ MeV}$$

The mass of a pion is $0.149m_p$. [2] If the pion produced by this interaction has a speed of $0.99c$, what is the energy of the tau neutrino?

$$E_{\bar{\nu}_\tau} = E_{\tau^-} - E_{\pi^-}$$

$$E_{\bar{\nu}_\tau} = \left(1 - \frac{u^2}{c^2}\right)^{-\frac{1}{2}} m_{\tau^-} c^2 - \left(1 - \frac{u^2}{c^2}\right)^{-\frac{1}{2}} m_{\pi^-} c^2$$

$$E_{\bar{\nu}_\tau} = (1 - 0.2^2)^{-\frac{1}{2}} \times 1776.86 - (1 - 0.99^2)^{-\frac{1}{2}} \times 0.149 \times \frac{1.67 \times 10^{-27} \times (3 \times 10^8)^2}{1.60 \times 10^{-13}}$$

$$E_{\bar{\nu}_\tau} = 821 \text{ MeV}$$

Accounting for relativistic effects, the energy and momentum of a particle can be related by:

$$E^2 - p^2 c^2 = m^2 c^4$$

As the tau neutrino is very light, we can effectively neglect the mass.

$$E^2 = p^2 c^2$$

$$p = \frac{E}{c}$$

$$p = \frac{821 \times 1.6 \times 10^{-13}}{3.0 \times 10^8}$$

$$p = 4.38 \times 10^{-19} \text{kg m s}^{-1}$$

What is the Koide mass ratio (equation 6) for the three leptons?

$$\text{Koide Mass Ratio} = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2}$$

$$\text{Koide Mass Ratio} = \frac{0.511 + 106 + 1780}{(\sqrt{0.511} + \sqrt{106} + \sqrt{1780})^2}$$

$$\text{Koide Mass Ratio} = 0.666 \approx \frac{2}{3}$$

What about the three heavy quarks?

$$\text{Koide Mass Ratio} = \frac{m_c + m_b + m_t}{(\sqrt{m_c} + \sqrt{m_b} + \sqrt{m_t})^2}$$

$$\text{Koide Mass Ratio} = \frac{0.511 + 106 + 1780}{(\sqrt{0.511} + \sqrt{106} + \sqrt{1780})^2}$$

$$\text{Koide Mass Ratio} = 0.666 \approx \frac{2}{3}$$

$$\text{Koide Mass Ratio} = \frac{m_c + m_b + m_t}{(\sqrt{m_c} + \sqrt{m_b} + \sqrt{m_t})^2}$$

$$\text{Koide Mass Ratio} = \frac{1270 + 4180 + 173000}{(\sqrt{1270} + \sqrt{4180} + \sqrt{173000})^2}$$

$$\text{Koide Mass Ratio} = 0.670 \approx \frac{2}{3}$$

Using the formula for the three charged leptons and heavy quarks yields a value close to $\frac{2}{3}$. Standard particle physics models cannot derive why these ratios result in simple fractions, and remains an unresolved mystery [8].

4 Compton Scattering

The Compton effect occurs when X-rays scatter off electrons, demonstrating wave-particle duality of photons. We can derive expressions for the recoil speed and angle of the electrons.

4.1 de Broglie Relation

The relativistic energy-momentum invariant relation for a particle of mass m is given by:

$$E^2 - p^2 c^2 = m^2 c^4 \quad (10)$$

For a photon, the rest mass is equal to zero.

$$E^2 = p^2 c^2$$

$$E = pc$$

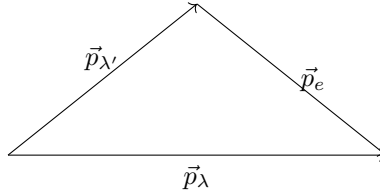
Using $E = \frac{hc}{\lambda}$:

$$p = \frac{h}{\lambda} \quad (11)$$

4.2 Electron Momentum from Conservation of Momentum

Conservation of momentum gives us the following equation:

$$\vec{p}_\lambda = \vec{p}_{\lambda'} + \vec{p}_e$$



We want to find the angle between λ and λ' , therefore we can use the cosine rule.

$$p_e^2 = p_\lambda^2 + p_{\lambda'}^2 - 2p_\lambda p_{\lambda'} \cos \theta \quad (12)$$

4.3 Electron Momentum from Energy Conservation

From conservation of total energy during the collision:

$$E_\lambda + E_{e,\text{rest}} = E_{\lambda'} + E_e$$

The incident photon energy is given by $E_\lambda = p_\lambda c$.

The scattered photon energy is given by $E_{\lambda'} = p_{\lambda'} c$.

The initial electron rest energy is given by $E_{e,\text{rest}} = m_e c^2$.

The total energy of the scattered electron is given by $E_e = \sqrt{p_e^2 c^2 + m_e^2 c^4}$.

Substituting these energy expressions into the conservation of energy equation:

$$p_\lambda c + m_e c^2 = p_{\lambda'} c + \sqrt{p_e^2 c^2 + m_e^2 c^4}$$

$$(p_\lambda - p_{\lambda'}) c + m_e c^2 = \sqrt{p_e^2 c^2 + m_e^2 c^4}$$

Dividing by the speed of light and squaring both sides:

$$(p_\lambda - p_{\lambda'}) + m_e c = \sqrt{p_e^2 + m_e^2 c^2}$$

$$(p_\lambda - p_{\lambda'})^2 + 2m_e c(p_\lambda - p_{\lambda'}) + m_e^2 c^2 = p_e^2 + m_e^2 c^2$$

$$p_e^2 = (p_\lambda - p_{\lambda'})^2 + 2m_e c(p_\lambda - p_{\lambda'}) \quad (13)$$

4.4 Wavelength Shift

We can set the two expressions for the squared electron momentum (equations 12 and 13) equal to each other.

$$p_\lambda^2 + p_{\lambda'}^2 - 2p_\lambda p_{\lambda'} \cos \theta = (p_\lambda - p_{\lambda'})^2 + 2m_e c(p_\lambda - p_{\lambda'})$$

$$p_\lambda^2 + p_{\lambda'}^2 - 2p_\lambda p_{\lambda'} \cos \theta = p_\lambda^2 - 2p_\lambda p_{\lambda'} + p_{\lambda'}^2 + 2m_e c(p_\lambda - p_{\lambda'})$$

$$-2p_\lambda p_{\lambda'} \cos \theta = -2p_\lambda p_{\lambda'} + 2m_e c(p_\lambda - p_{\lambda'})$$

$$2p_\lambda p_{\lambda'} (1 - \cos \theta) = 2m_e c(p_\lambda - p_{\lambda'})$$

Substituting the de Broglie relations (equation 11) for the initial and scattered photon momentum:

$$\frac{h}{\lambda} \frac{h}{\lambda'} (1 - \cos \theta) = m_e c \left(\frac{h}{\lambda} - \frac{h}{\lambda'} \right)$$

$$\frac{h^2}{\lambda \lambda'} (1 - \cos \theta) = m_e c h \left(\frac{\lambda' - \lambda}{\lambda \lambda'} \right)$$

Multiplying both sides by the product of wavelengths and dividing by Planck constant:

$$h(1 - \cos \theta) = m_e c(\lambda' - \lambda)$$

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e c}(1 - \cos \theta) \quad (14)$$

4.5 Angle for Largest Shift and Photon Energy for 10% Shift

The shift in wavelength is maximized when $\cos \theta = -1$. This happens when $\theta = 180^\circ$

The maximum possible wavelength shift is therefore:

$$\Delta\lambda_{\max} = \frac{2h}{m_e c}$$

For a ten percent shift relative to the incident wavelength $\Delta\lambda_{\max} = 0.10\lambda$

$$\frac{2h}{m_e c} = 0.10\lambda$$

$$\lambda = \frac{20h}{m_e c}$$

Substituting $E = \frac{hc}{\lambda}$:

$$E = \frac{hc}{\frac{20h}{m_e c}}$$

$$E = \frac{m_e c^2}{20}$$

$$E = \frac{9.1 \times 10^{-31} \times (3.0 \times 10^8)^2}{1.6 \times 10^{-16} \times 20} = 26 \text{ keV}$$

4.6 Recoil Speed, Angle, and Momentum Expressions

The following equation gives the relativistic momentum of a recoiling electron [2].

$$p_e = \gamma m_e v \quad (15)$$

Where the Lorentz factor γ is defined as:

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}} \quad (16)$$

Expressing the normalized momentum ratio:

$$\frac{p_e}{m_e c} = \frac{v}{c} \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$$

Squaring both sides and taking the reciprocal:

$$\left(\frac{p_e}{m_e c}\right)^2 = \frac{v^2}{c^2} \left(1 - \frac{v^2}{c^2}\right)^{-1}$$

Taking the reciprocal:

$$\begin{aligned} \left(\frac{m_e c}{p_e}\right)^2 &= \frac{c^2}{v^2} \left(1 - \frac{v^2}{c^2}\right) \\ \left(\frac{m_e c}{p_e}\right)^2 &= \frac{c^2}{v^2} - 1 \\ \frac{c^2}{v^2} &= 1 + \left(\frac{m_e c}{p_e}\right)^2 \end{aligned}$$

Inverting and taking the square root yields the recoil speed.

$$v = c \left(1 + \left(\frac{m_e c}{p_e}\right)^2\right)^{-\frac{1}{2}}$$

We can apply momentum conservation along the two axes for the recoil angle.

$$p_e \cos \phi = p_\lambda - p_{\lambda'} \cos \theta$$

$$p_e \sin \phi = p_{\lambda'} \sin \theta$$

Dividing the perpendicular component by the parallel component:

$$\tan \phi = \frac{p_{\lambda'} \sin \theta}{p_\lambda - p_{\lambda'} \cos \theta}$$

Dividing both the numerator and denominator by the initial photon momentum:

$$\tan \phi = \frac{\frac{p_{\lambda'}}{p_\lambda} \sin \theta}{1 - \frac{p_{\lambda'}}{p_\lambda} \cos \theta}$$

Using the equation for the wavelength shift (equation 14) we can find the wavelength ratio.

$$\frac{\lambda'}{\lambda} = 1 + \frac{h}{m_e c \lambda} (1 - \cos \theta)$$

We can substitute this into the formula for the recoil angle.

$$\tan \phi = \frac{\sin \theta}{1 + \frac{h}{m_e c \lambda}(1 - \cos \theta) - \cos \theta} \quad (17)$$

4.7 Maximum Electron Recoil Speed

As mentioned before, maximum energy transfer occurs when $\theta = 180^\circ$. The scattered wavelength (equation 14) becomes:

$$\lambda' = \lambda + \frac{2h}{m_e c}$$

The equation for conservation of energy is:

$$E_\lambda + m_e c^2 = E_{\lambda'} + E_e$$

Initial photon energy is: $E_\lambda = \frac{hc}{\lambda}$.

Scattered photon energy is: $E_{\lambda'} = \frac{hc}{\lambda'}$.

Therefore, the total energy of the recoiling electron is:

$$E_e = m_e c^2 + \frac{hc}{\lambda} - \frac{hc}{\lambda'}$$

The total energy is related to the Lorentz factor (equation 15): $E_e = \gamma m_e c^2$.

The Lorentz factor for the recoil electron is:

$$\gamma = 1 + \frac{\frac{hc}{\lambda} - \frac{hc}{\lambda'}}{m_e c^2}$$

Relating recoil speed to the Lorentz factor:

$$\frac{v}{c} = \sqrt{1 - \frac{1}{\gamma^2}}$$

The inverse Lorentz factor is:

$$\frac{1}{\gamma} = \frac{m_e c^2}{m_e c^2 + \frac{hc}{\lambda} - \frac{hc}{\lambda'}}$$

Finally, substituting the inverse Lorentz factor into the speed formula yields:

$$v = c \sqrt{1 - \left(\frac{m_e c^2}{m_e c^2 + \frac{hc}{\lambda} - \frac{hc}{\lambda'}} \right)^2} \quad (18)$$

4.8 Plots of Fractional Wavelength and Electron Recoil Speed

We used equation 14 to plot fractional wavelength and equation 18 to plot electron recoil speed against photon scattering angle against photon scattering angle. Each line represents a different energy of the incident X-ray photon.

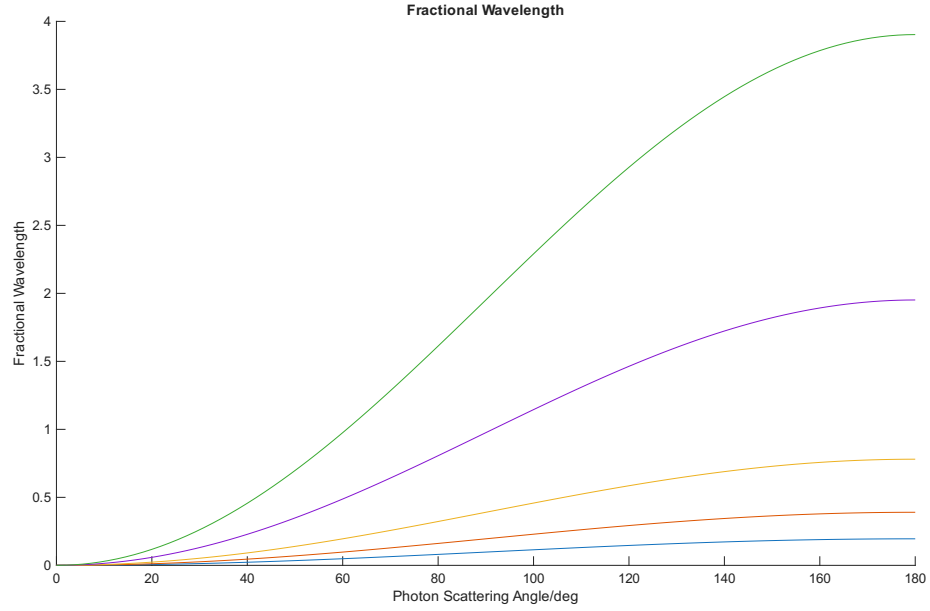


Figure 6: Fractional Wavelength against Photon Scattering Angle

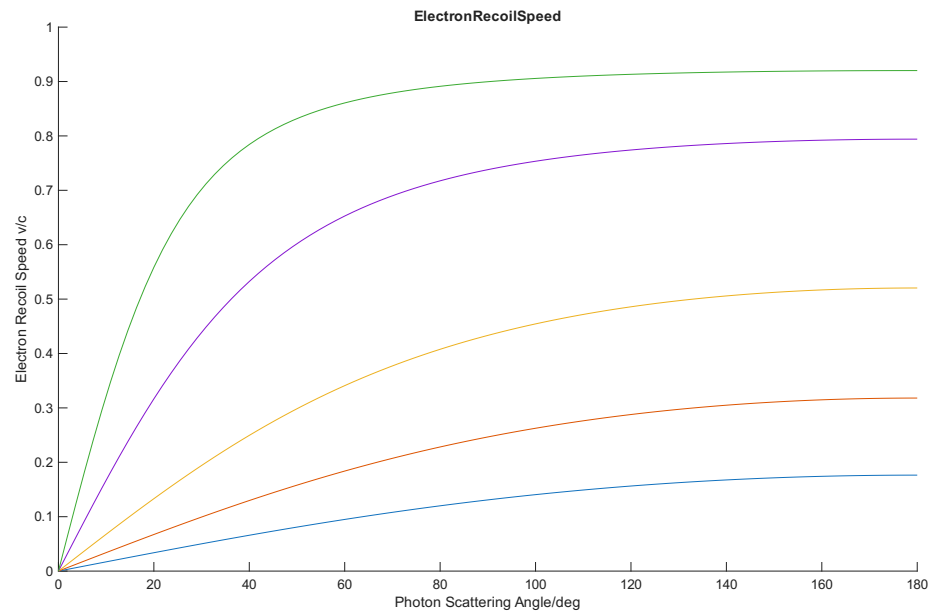


Figure 7: Electron Recoil Speed against Photon Scattering Angle

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A MATLAB Code for Plots

```
%% Constants
clc
e = 1.6e-19;
c = 3.0e8;
h = 6.626e-34;
m = 9.11e-31;
E = [50000*e,100000*e,200000*e,500000*e,1000000*e];
lambda = (h*c)./E;
theta = linspace(0,pi,100);
theta_deg = theta./pi*180;

%% Fractional Wavelength
constant = h/(m*c);
delta_lambda = constant.*(1-cos(theta));
figure
hold on
for i = 1:5
    fractional_wavelength = delta_lambda./lambda(i);
    plot(theta_deg,fractional_wavelength)
end
xlabel("Photon Scattering Angle/deg")
ylabel("Fractional Wavelength")
title("Fractional Wavelength")
hold off

%% Electron Recoil Speed
figure
hold on
for j = 1:5
    new_lambda = delta_lambda + lambda(j);
    num = c.*sqrt(1-((m*c^2);
    den = h*c/lambda(j)-h*c./new_lambda+m*c^2)).^2;
    v = num./den;
    plot(theta_deg,v/c)
end
xlabel("Photon Scattering Angle/deg")
ylabel("Electron Recoil Speed v/c")
title("ElectronRecoilSpeed")
hold off
```